Lesson plan · for the teacher

Build a rain gauge

IoT · Level 1 · Lesson 3

Time
40 min

Materials
One clear plastic bottle with straight, smooth sides per group; Scissors or a knife for the teacher only; tape; Small stones, water, a ruler and a marker or pen; Paper strips for a millimetre scale

Learning goals

- Explain that rainfall is measured in millimetres of depth.
- Build a rain gauge from a bottle, with a zero line, a millimetre scale and a funnel.
- Choose a good place for the gauge and agree a daily reading routine.

Activity: Make a bottle rain gauge: Build

1. Warm-up (5 min): Ask: “Did it rain a lot or a little last time?” Let pupils disagree. Write their words on the board — “a lot”, “a bit”, “heavy”.
2. Teach (7 min): Explain millimetres of rain: the depth on flat ground. Show a wide and a narrow straight-sided container and ask: in the same rain, which fills deeper? Explain that both reach the same depth.
3. Build (15 min): Groups add stones and water just to cover them, mark the 0 line, and tape on a paper scale marked every millimetre from a ruler. Then they turn the cut top into the bottle as a funnel and tape it.
4. Place it (8 min): Walk to the spots you chose. Groups judge each one: open sky? Far from roofs and trees? Safe from feet and animals? Place the gauge in the best spot.
5. Wrap-up (5 min): Back in class, draw a reading rota and a table on the board: date, time, rain (mm). Exit ticket: one rule for reading the gauge.

Before class, cut the top off each bottle where the sides start to curve, and tape the cut edges. Check that each bottle’s sides are straight. Walk the school grounds and pick two or three open spots to discuss.

Discussion

- Is your 0 line level?
- Does every millimetre on your scale match the ruler?
- Is there anything above the gauge that could block rain or drip water in?

Exit ticket

- One rule for reading the rain gauge, and why it matters.

Try it in class tomorrow
Build a class rain gauge: Bring a clear, straight-sided bottle. Cut off the top yourself, tape the edge, and let pupils add stones, water to a 0 line, a paper mm scale and the funnel. Agree an open spot and a daily reading time, and draw a rota on the board.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Your school has three possible spots for the rain gauge. Which is best?**

- ◯ Beside the classroom wall, under the roof edge
- ◯ Under the big tree by the gate, out of the wind
- ◯ In the middle of the open school field

2 **Put the steps for building and using the class rain gauge in order.**

- ☐ Set it in open ground, away from trees
- ☐ An adult cuts off the bottle's top
- ☐ Read it at the same time each day
- ☐ Write it down, then empty it back to 0
- ☐ Add a mm scale and the funnel
- ☐ Add stones and water; mark the 0 line

3 **A wider bottle will show more millimetres of rain than a narrow one.**

◯ True ◯ False

4 **Match each part or habit to its reason.**

Straight sides	• • • •	Each reading covers one full day
Upside-down top as a funnel	• • • •	Stop it blowing over
Stones in the bottom	• • • •	No rain is counted twice
Same time each day	• • • •	Every millimetre on the scale is true
Emptying after reading	• • • •	Guides rain in, keeps most leaves out

5 **Complete the sentence.**

Rainfall is measured in _____ : the _____ the rain would reach on flat ground.

Word bank: litres · millimetres · weight · depth

Answers · for the teacher

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1 **In the middle of the open school field**

In the open field, nothing blocks the rain or drips extra water in, so the reading is honest.

2 **Top to bottom: 4, 1, 5, 6, 3, 2**

Build, then place, then read. Writing the reading down before you empty the gauge means no reading is lost — and no rain is counted twice.

3 **False**

The wider bottle catches more water, but with straight sides the water stands at the same depth.

4 **Straight sides — Every millimetre on the scale is true**
Upside-down top as a funnel — Guides rain in, keeps most leaves out
Stones in the bottom — Stop it blowing over
Same time each day — Each reading covers one full day
Emptying after reading — No rain is counted twice

Every step of the build has a reason. When pupils can say the reason, they can fix a gauge that goes wrong.

5 **Rainfall is measured in millimetres: the depth the rain would reach on flat ground.**

Rain gauges measure depth. 10 mm of rain means water 10 mm deep, if none ran off, soaked in or dried up.

For you to think about

Where in your school could a rain gauge stand in the open, safe from animals and balls? Who would read it each day?